

Strategic Reviews

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Abstract. The impact of product reviews on consumer purchasing behavior is empirically well documented. This can create perverse incentives for firms to offer reviewers side payments (“bribes”) in exchange for biased reviews for their products. The presence of bribes distorts the information in reviews and leads to detrimental effects on consumer utility. This paper builds a two-sided reputation model where a reviewer can inflate her reviews in exchange for bribes. If the reviewer accepts bribes and misrepresents her reviews, then she builds her reputation as an inaccurate reviewer and makes consumers less likely to follow her recommendations. This in turn makes firms no longer interested in offering her a bribe. Can the reviewer retain influence over consumers’ purchasing decisions while simultaneously accepting bribes and misrepresenting her reviews? We provide a characterization of the environments that allow this kind of manipulation and show that regulatory policies that aim to reduce bribes can lead to undesirable outcomes. Finally, we show that eliminating bribes from the market can increase the welfare of all market participants, *even* for those firms who would have otherwise bribed in exchange for more favorable reviews.

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1. Introduction

Consumers often make purchasing decisions by relying on reviews. This can make it attractive for firms to offer reviewers side payments—“bribes”—in exchange for inflated ratings for their products. A recent example is the case of PewDiePie, a popular YouTube gamer who was at the center of a settlement between the Federal Trade Commission (FTC) and Warner Bros, after the latter was found to have offered thousands of dollars in compensation to reviewers to “promote the game in a positive way and not to disclose any bugs or glitches they found” (FTC 2016b).

This paper focuses on influential reviewers like the previous example. We develop a two-sided reputation model where a reviewer with a *private* ability type is situated between consumers and firms. The reviewer is either low or high skilled, and this type determines her accuracy in evaluating the quality of the firms’ products. Consumers observe the reviewer over time and learn how accurate she is by comparing her reviews to the ex post realized qualities of the products. A more accurate reviewer becomes more influential with consumers, who rely on her reviews to make purchasing decisions. The reviewer obtains utility from this influence, for example, through direct support from consumers on platforms like Patreon or from advertising revenue on dissemination channels like YouTube.

An influential reviewer is also an appealing target for bribing by firms. In our model, firms are either honest (do not bribe) or strategic (offer bribes if it is profitable), and this attribute is unobservable to consumers. If the reviewer accepts bribes and starts misrepresenting her reviews, her accuracy declines, and her influence diminishes. This in turn makes firms no longer interested in bribing her. We identify conditions under which the equilibria of this game admit situations where the reviewer consistently accepts bribes and misleads consumers. We then use our model to discuss the implications to the utility of all players in the market: consumers, firms, and the reviewer.

1.1. Contribution and Overview of Results

Here is a summary of our main results.

1.1.1. Characterization. We first show in Theorem 1 that the previous game admits at most two equilibria. In the *separating* equilibrium, both the low and high type reviewers report truthfully, no bribes are offered, and all firms get fair reviews. In the *pooling* equilibrium, the low type reviewer reports truthfully but the high type mimics a low type and posts less accurate reviews. Of the many ways in which she can pretend to be a low type, she chooses a particular one: she injects positive bias into the reviews of firms that

bribe her and negative bias into the reviews of firms that do not. In this equilibrium, all the strategic firms bribe and get inflated reviews, whereas all the honest firms get depressed reviews. Consumers do not observe which firms bribe, so they cannot tell if the reviewer is providing bias to some firms over others or is simply inaccurate. As a result, consumers are consistently biased toward the products of bribing firms.

The separating equilibrium *always* exists, but the pooling equilibrium may not. If the pooling equilibrium does not exist, we say the environment is *bribe proof*. We provide detailed conditions under which the pooling equilibrium (and bribes) can exist.

1.1.2. Comparative Statics. Next, we use our model to deliver insights about how the different parameters of the environment affect bribes. Theorem 2 shows that an increase in the proportion of honest firms can lead to *lower* consumer utility. This is because the pooling equilibrium only exists if the high type reviewer can credibly play like a low type. If she biases the reviews of bribing firms up, she needs to bias other firms down so that her overall accuracy is consistent with that of the low type reviewer. The presence of honest firms in the market allows her to do this and maintain a facade of being impartial and fair. A potential implication is that regulations that aim to reduce the proportion of bribing firms in the market (e.g., through the threat of auditing and fines) can end up making it easier for the reviewer to accept bribes and manipulate her reviews. These nonmonotone effects of regulation on consumer welfare have recently been observed in the empirical work of Ershov and Mitchell (2020).

Consumer utility is also affected by the *ex ante* information that the market has about the skill of the reviewer. Propositions 2 and 3 illuminate an important counterintuitive finding: increasing the expected skill of the reviewer may not necessarily lead to better outcomes for consumers. This is because more skilled reviewers have more leeway in inflating their reviews, and this makes the prospect of bribing them more appealing to firms. Thus, this improvement in reviewer quality might allow both reviewers and firms to extract greater surplus through bribes and inflated reviews at the cost of hurting consumers.

1.1.3. Total Welfare. Finally, Proposition 4 gives a mild condition under which bribe-proof environments improve the total welfare of *all* market participants (consumers, reviewer, and firms). Interestingly, it can be the case that the bribing firms themselves would prefer a world without bribes. In these instances, the reviewer accepts bribes to not hurt the firm by giving it a poor review. The bribe that the firm pays to the reviewer becomes a form of extortion or “protection

money”: it does not benefit the firm in any way, but rather helps it avoid a worse fate.

1.1.4. Commitment vs. Fully Dynamic Model. Throughout the paper, we present the previous findings using a simplified game-theoretic model that assumes the reviewer commits to playing according to a certain (low or high) type. This commitment assumption greatly simplifies the exposition, but importantly, we show through detailed analyses and equilibrium refinement choices that commitment emerges as the unique equilibrium of the full dynamic interaction between players. For the interested reader, the details and analysis of the full dynamic model are included in Online Appendix B.

1.2. Related Literature

The effect of reviews on consumer purchasing behavior is empirically well documented in Chevalier and Mayzlin (2006), Senecal and Nantel (2004), and Dellarocas et al. (2007). The phenomenon of influential online reviewers is relatively recent but bears some resemblance to earlier work that documents the effect of movie critics on viewership (Reinstein and Snyder 2005). There is also an emerging literature on how consumers learn from reviews (Acemoglu et al. 2017, Ifrach et al. 2019), which adopts a Bayesian sequential learning approach where consumers leave reviews of products, and these reviews are used by future consumers to determine the quality of the product. Chen and Papanastasiou (2021) study how a firm can manipulate this sequential learning process through planting an initial “fake” purchase, whereas Dellarocas (2006) is an earlier work that studies a game where firms try to manipulate opinions in online forums and shows when such manipulation is beneficial to consumers or firms. These papers are part of a broader literature that examines manipulation of agents in various online settings (Candogan and Drakopoulos 2020, Belavina et al. 2020, Papanastasiou 2020, Mostagir et al. 2022).¹

Unlike the previously mentioned literature, our paper conceives of the reviewer as a proper agent acting strategically to further her own interests, and therefore, we model all three types of actors. This relates our paper to the literature where an expert communicates information to a buyer on behalf of a seller. Lizzeri (1999) studies the incentives of a monopolistic expert to accurately reveal information and Inderst and Ottaviani (2012) study how firms can compete for market share by trying to influence the expert through kickbacks. In a similar vein, the recent paper of Fainmesser and Galeotti (2021) studies competition between influencers who choose how much sponsored versus organic content to provide. The equilibria of our simplified commitment model lead to different findings from the previous papers. This is a consequence of the fact that the commitment model is the reduced-form outcome of the repeated interactions

that allow both sides of the market to learn about the unknown type of the reviewer: an element that is absent in the aforementioned papers. Mitchell (2021) models the dynamic relationship between a reviewer and a consumer but does not model firms, which leads to a different incentive structure and different results from the three-player model.

Finally, our paper is related to the recent work on two-sided reputation, as in Bar-Isaac and Deb (2014) and Bouvard and Levy (2017), where, in the latter, an agent is faced with the problem of providing certification for sellers of different types. Our work shares some similarities, but there are also several important distinctions. Preferences in our paper are aligned (both consumers and firms prefer a high type to a low type reviewer) but are conflicting in Bouvard and Levy (2017). In addition, the lack of commitment in Bouvard and Levy (2017), which underscores the fact that commitment arises from the repetition in the dynamic model, leads to different outcomes because the reviewer in that paper can misrepresent herself in the first period to take advantage in the second (and final) period. Finally, there are also differences that stem from that fact that, although firms elect to be certified (and therefore induce a signaling game), they cannot choose not to be reviewed, as per the Consumer Review Fairness Act of 2016 (FTC 2016a). Durbin and Iyer (2009) consider a model where an advisor (reviewer in our setting) is more concerned about her reputation for honesty rather than her accuracy, which gives rise to different dynamics as the authors themselves note in their paper.²

The next section demonstrates the main ideas of the paper in a reduced-form example. Our formal model and equilibrium characterization are presented in Section 3. Section 4 uses this model to describe what environments admit bribes and examine the effect of the different parameters on consumer utility and total welfare. Section 5 concludes the paper. All proofs are in Appendix A. Interested readers can find the equilibrium analysis that allows us to reduce the full dynamic model to the commitment model used throughout the paper in Online Appendix B. An optional discussion that connects the main model of the paper to the full dynamic model is in Section 3.5.

2. Reduced-Form Example

We provide intuition for our results through a stylized example. A reviewer repeatedly interacts with a mass of consumers for whom she writes reviews which inform purchasing decisions. Firms arrive sequentially in discrete time, and some of these firms are “friends” of the reviewer. The reviewer would like to help her friends out but is impartial to those who are not her friends. We are interested in whether the reviewer can repeatedly

steer the *Bayesian* consumers toward products sold by her friends, without those consumers being able to tell that this is what she is doing.

The setup is as follows. Time is discrete $t = 1, 2, \dots$. Firm t arrives at time t and sells a product whose quality q_t is distributed as $\mathcal{N}(0, 1)$. The reviewer samples the product and observes a signal $s_t = q_t + \varepsilon_t$, where $\varepsilon_t \sim \mathcal{N}(0, \sigma_\omega^2)$ and is i.i.d across time and independent of product quality. The reviewer’s private type/skill ω is either High ($\sigma_H = 1$) or Low ($\sigma_L = 2$), and both types are equally likely. After observing her signal, the reviewer posts a review r_t , which may or may not be the same as s_t . A reviewer cannot pretend to be more accurate than she actually is, so type H can pretend to be type L, but the opposite is not possible. If the reviewer always tells the truth regardless of her type, then type L reviews are distributed as $r_t^L \sim \mathcal{N}(0, 5)$, and type H reviews are distributed as $r_t^H \sim \mathcal{N}(0, 2)$.

The true quality q_t is eventually revealed to every one before the next period. Consumers are aware of the possible reviewer types and the probabilities with which they occur, so they can learn the reviewer’s skill by comparing her reviews with the revealed qualities over time. Consumers are also aware that some firms might be friends of the reviewer, *but they do not know which firms are friends and which are not*. If consumers believe that the reviewer is intentionally misrepresenting her reviews, they stop listening to her forever, which is an outcome that she wants to avoid. If the reviewer systematically boosts the reviews of her friends, then she will be behaving in a way that is inconsistent with being either type L or type H, and this will trigger consumers to ignore her.

Before they make their purchasing decisions, consumers optimally extract information about the quality q_t from the review r_t . They do this through adjusting the review by the precision of what they *believe* the reviewer’s type is (this is the standard inverse-variance weighting formula, as in, e.g., Cochran (1954), that we also discuss in Section 3). In particular, if consumers believe that the reviewer’s type is $\hat{\omega} \in \{L, H\}$, then the expected quality given the review r_t and the type $\hat{\omega}$ is given by

$$\mathbb{E}[q_t | r_t, \hat{\omega}] = \frac{r_t}{1 + \sigma_{\hat{\omega}}^2}.$$

Now suppose that in this sequence of firms, with probability $1/2$ a firm is a friend of the reviewer and with probability $1/2$ it is not. Let Z_t be the random variable indicating whether firm t is a friend ($Z_t = +1$) or not ($Z_t = -1$). Assume that the reviewer is indeed type H but that no one else knows this. As mentioned, if she reports truthfully, then her review for any firm (friend or not) is distributed as $r_t \sim \mathcal{N}(0, 2)$. Consumers, having learned that the reviewer is type H, would believe that the expected quality given a review r_t is equal to $\mathbb{E}[q_t | r_t, \hat{\omega} = H] = r_t/2$ and therefore the

average quality belief for *any* firm is

$$\mathbb{E}_{q_t}[\mathbb{E}[r_t | q_t]/2] = 0.$$

As expected, when the reviewer behaves truthfully, consumers do not prefer friends to nonfriends, as both types get the same reviews on average. Instead of reporting truthfully however, the reviewer can pretend to be type L. She can do this in a specific way that benefits her friends as follows: she takes her signal s_t , injects noise ε'_t , and writes a review $r'_t = s_t + \varepsilon'_t$, where $\varepsilon'_t = |X_t| \cdot Z_t$ and $X_t \sim \mathcal{N}(0, \sigma_L^2 - \sigma_H^2) = \mathcal{N}(0, 3)$. It is easy to see that $\varepsilon'_t \perp s_t$ and $\varepsilon'_t \sim \mathcal{N}(0, 3)$. Therefore, $r'_t \sim \mathcal{N}(0, 5)$. Importantly, r'_t is statistically unidentifiable from the random variable $r_t^L \sim \mathcal{N}(0, 5)$, which is the distribution of the reviews written by a truth-telling reviewer of type L. Consumers cannot distinguish whether the reviewer is indeed type L or is a type H pretending to be type L (and biasing her friends in the process). All they can observe is a review distribution that looks like the bottom curve in Figure 1.

Now, notice that, although the reviews over all firms are normally distributed, the reviews for friends are distributed as in the top-left curve of Figure 1. The average review for friends of quality q_t is therefore

$$\mathbb{E}[r'_t | q_t] = \mathbb{E}[s_t + \varepsilon'_t | q_t] = \mathbb{E}[s_t + |X_t| | q_t] = q_t + \sqrt{\frac{6}{\pi}},$$

whereas the reviews for nonfriends are distributed as in the right curve of Figure 1. The average review for nonfriends of quality q_t is

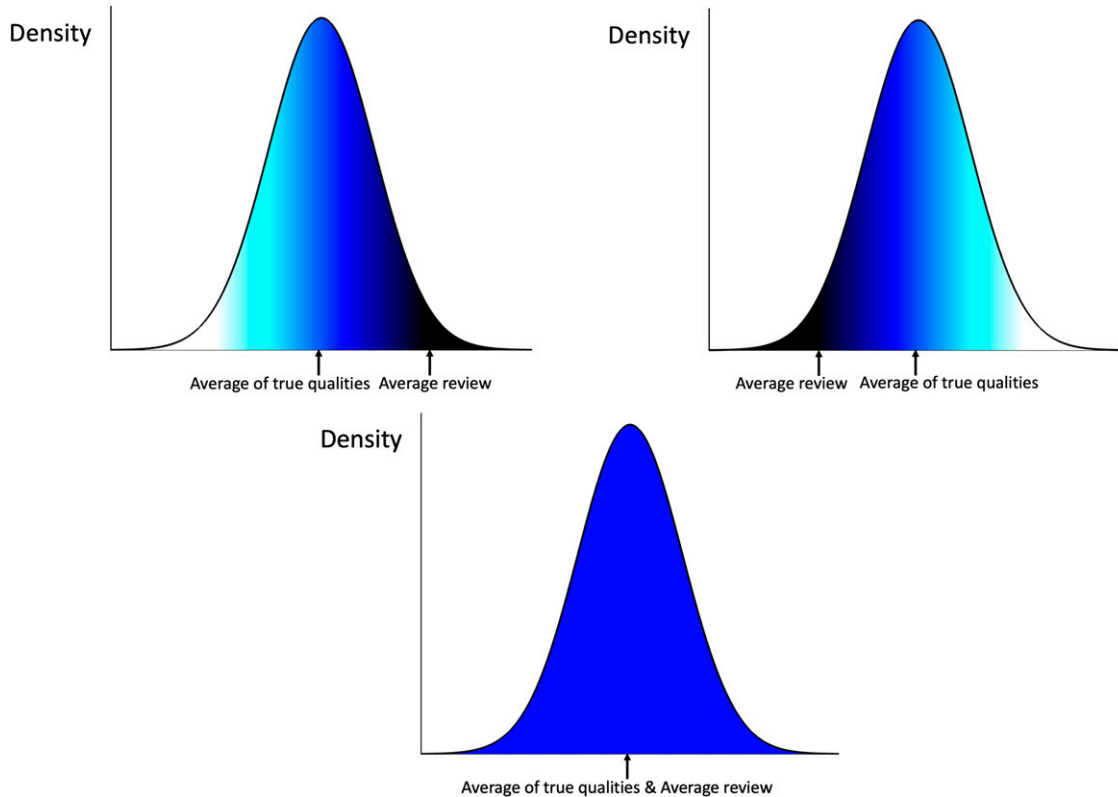
$$\mathbb{E}[r'_t | q_t] = \mathbb{E}[s_t + \varepsilon'_t | q_t] = \mathbb{E}[s_t - |X_t| | q_t] = q_t - \sqrt{\frac{6}{\pi}}.$$

From the consumer's perspective, having learned that the reviewer is playing as type L, she believes the expected quality given a review r'_t is equal to $\mathbb{E}[q_t | r'_t, \hat{\omega} = L] = \frac{r'_t}{1 + \sigma_L^2} = r'_t/5$. This means that the average quality belief for friends is

$$\mathbb{E}_{q_t}[\mathbb{E}[r'_t | q_t]/5] = \sqrt{\frac{6}{25\pi}}.$$

Similarly, consumers (on average) believe the quality of nonfriends to be $-\sqrt{\frac{6}{25\pi}}$. Because consumers cannot tell which firms are friends or not friends, they will inherently be biased toward friends of the reviewer, unless the reviewer's true type is actually L. $\square$

Figure 1. (Color online) Distribution of Reviews (and True Quality) for Friends, non-Friends, and Overall.



Notes. The top two curves display the review distributions for friends (left curve) and nonfriends (right curve). The darker the shading, the more likely the review is to come from this part of the curve (compared with a fair review). Consumers cannot tell reviews are coming from the top curves; instead, they can only observe the bottom distribution, which shows no bias.

The rest of the paper generalizes the previous example. In particular, being a friend arises endogenously through those firms who choose to offer the reviewer a bribe (and hence become friends). In the model, firms are myopic and bribes are not contractible, so it is possible that the reviewer collects the bribe and not follow through with a biased review. Firms, anticipating this, might not want to offer a bribe to begin with. However, we show that this game can sometimes admit an equilibrium where firms offer bribes in exchange for bias from reviewers. We then use our model to deliver additional insights about how consumer utility changes as a function of the parameters of the problem.

3. Model

We consider the following dynamic model. Firms are agents that produce and sell a product to a unit mass of consumers on $[0, 1]$. Firms arrive sequentially in discrete time and market their product to consumers through a reviewer. The reviewer broadcasts her experiences of products to consumers and might do so in a strategic way to influence purchasing decisions.

3.1. Timing

At $t = 0$, the skill of the reviewer, ω , is drawn from a known Bernoulli distribution over the finite type space $\Omega = \{L, H\}$ (for Low and High-skill, respectively), where $\omega = H$ is drawn with probability p . These skill types correspond to the reviewer's precision, σ_ω , in assessing the firm's product, with $\sigma_H < \sigma_L$. The reviewer knows her actual type ω , but firms and consumers only know the type distribution.

The following sequence of events happens at every time $t \geq 1$:

(a) Firm t arrives with a product of random quality q_t , which is drawn from a standard normal distribution $q_t \sim \mathcal{N}(0, 1)$. With probability θ , the firm is honest and offers no bribe. With probability $1 - \theta$ the firm is strategic and decides to either **Bribe** (by offering $b_t > 0$) or **Not Bribe** (by offering $b_t = 0$).

(b) The reviewer samples the product and receives an unbiased, noisy signal $s_t = q_t + \varepsilon_t$ of the quality, where $\varepsilon_t \sim \mathcal{N}(0, \sigma_\omega^2)$ and is independent and identically distributed (i.i.d.) across time and independent of q_t . The reviewer posts a review $r_t \in \mathbb{R}$ of the product, where r_t is not necessarily equal to s_t .

(c) Every consumer $i \in [0, 1]$ observes the review r_t and then, based on r_t , chooses whether to purchase the good $x_{i,t} \in \{0, 1\}$ at unit price.

We assume the true quality q_t is revealed through a public signal after each period and that the history of reviews $\{r_1, \dots, r_{t-1}\}$ and true qualities $\{q_1, \dots, q_{t-1}\}$ is publicly known. The history of bribes $\{b_1, \dots, b_{t-1}\}$, on the other hand, is private to the reviewer and is not

visible to consumers or firms. We refer to the previous model as the *full dynamic model*.

3.2. Actions

Throughout the paper, we focus on a version of that model where all players have binary actions. Consumers either buy ($x_i = 1$) or not ($x_i = 0$), strategic firms either offer bribe $b_t = b^*$ or $b_t = 0$,³ and the reviewer plays according to the commitment model we outline later. This is without loss of generality: For the interested reader, the connection to the more general strategy spaces of the full dynamic model is discussed in Section 3.5 and the formal equivalence is shown in Online Appendix B.

3.2.1. Commitment. We assume the reviewer commits at time 0 to playing an *effective* type $\hat{\omega} \in \{L, H\}$. This effective type is possibly distinct from her true type ω . A low-skilled reviewer ($\omega = L$) must choose $\hat{\omega} = L$, because she cannot pretend to be more skilled than she actually is. The more interesting case is the high-skilled reviewer ($\omega = H$) who can choose either $\hat{\omega} = H$ or $\hat{\omega} = L$.⁴ If she chooses the latter, then her reports are less accurate compared with the case when she reports truthfully.

Firms and consumers observe the choice of effective type $\hat{\omega}$. Although we take this observation as an assumption, the fact that a reviewer commits to an effective type $\hat{\omega} \in \{L, H\}$ means that agents will learn this type (i.e., how she actually plays over time) after observing enough history of past play. For this reason, we will use *effective type* and *reputation* interchangeably when we refer to $\hat{\omega}$.

If the type H reviewer commits to $\hat{\omega} = H$, she writes (truthful) reviews $r_t = s_t$. However, if she commits to $\hat{\omega} = L$, then she distorts her reviews in a way that makes her statistically indistinguishable from a true type L. The set of functions mapping s_t to r_t such that r_t is statistically identical to a type L reviewer is not unique. We focus on a specific mapping, which is a generalization of the strategy in Section 2. In this mapping, which is parameterized by the proportion of honest firms θ , the type H reviewer writes reviews $r_t = s_t + \varepsilon'_t$ in the following way:

- If the firm chooses **Not Bribe**, then the reviewer adds noise ε'_t from the bottom θ percentile of the $\mathcal{N}(0, \sigma_L^2 - \sigma_H^2)$ distribution.
- If the firm chooses **Bribe**, then the reviewer adds noise ε'_t from the top $(1 - \theta)$ percentile of the $\mathcal{N}(0, \sigma_L^2 - \sigma_H^2)$ distribution.

Under the previous mapping, bribing firms are given the most upward bias possible and nonbribing firms are given the most downward bias, subject to the reviewer having to appear as a type L.⁵ This leads to the biggest gain from bribing out of all maps from s_t to r_t and therefore provides the most incentive for

firms to bribe.⁶ Hence, to determine whether bribes can exist in the system, it is enough to check if firms are willing to bribe under this mapping.

3.3. Payoffs

We now describe the payoffs of the three types of agents.

3.3.1. Consumers. Recall there is a continuum of consumers on $[0, 1]$ with consumer i having outside option $\phi(i)$. For simplicity, we make the following assumption.

Assumption 1. The term $\phi(i)$ is increasing in i , with $\lim_{i \rightarrow 0} \phi(i) = -\infty$ and $\lim_{i \rightarrow 1} \phi(i) = \infty$.

Assumption 1 guarantees that for all t , a positive fraction of consumers purchase the time t product and a positive fraction do not. In particular, the map $\phi^{-1}(\alpha)$ is well defined for any $\alpha \in \mathbb{R}$ and corresponds to the measure of consumers with $\phi(i) \leq \alpha$. Our results hold for any outside option distribution that satisfies Assumption 1.

Consumers who purchase the product experience the true quality q_t .⁷ Consumers are myopic and maximize their current-period utility given the posted review r_t and the reputation of the reviewer $\hat{\omega}$ (which provides information about the accuracy of the reviewer); that is, consumer i chooses her consumption $x_{i,t} \in \{0, 1\}$ according to

$$x_{i,t}^*(r_t, \hat{\omega}) = \arg \max_{x_{i,t} \in \{0, 1\}} \mathbb{E}[(q_t - \phi_i)x_{i,t} | r_t, \hat{\omega}].$$

It is easy to see that each consumer will use a cutoff strategy $x_{i,t}^*(r_t, \hat{\omega}) = 1$ if and only if (iff) $\mathbb{E}[q_t | r_t, \hat{\omega}] \geq \phi_i$. Therefore, we can define $X_t^*(r_t, \hat{\omega}) = \mu(\{i : x_{i,t}^*(r_t, \hat{\omega}) = 1\})$, where μ is the Lebesgue measure, as the total consumption of the product. Note that $X_t^*(r_t, \hat{\omega}) = \phi^{-1}(\mathbb{E}[q_t | r_t, \hat{\omega}])$.

3.3.2. Reviewer. The reviewer is a far-sighted agent who discounts her payoff over time at rate $\delta \in (0, 1)$. The reviewer's payoff reflects a desire to obtain (i) influence on consumer purchasing decisions and (ii) any potential bribes she can solicit.

(i) *Influence:* The influence I_t of the reviewer measures how much consumption reacts to her review on average. Observe that $\phi^{-1}(0)$ is the consumption when no review is written and $\phi^{-1}(\mathbb{E}[q_\tau | r_\tau, \hat{\omega}])$ is the consumption when review r_τ is written at time τ , then

$$I_t = \frac{1}{t} \sum_{\tau=1}^t (\phi^{-1}(\mathbb{E}[q_\tau | r_\tau, \hat{\omega}]) - \phi^{-1}(0))^2. \quad (1)$$

The previous equation implies that the influence of the reviewer depends on how she was able to change average consumption patterns in the past.

We assume that the reviewer cares about her influence with propensity $\beta \geq 0$. The case $\beta > 0$ accounts for the possibility that the reviewer derives direct revenue

and/or utility from her influence. This may come, for example, from advertisements on her dissemination channel or direct payments from her patrons.

(ii) *Bribes:* Reviewers may accept (noncontractual) bribes b_t from firms. These bribes are only observed by the reviewer and the bribing firm.

Reviewers maximize their long-term average payoff:

$$V = (1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} (\beta \cdot I_t + b_t).$$

3.3.3. Firms. Recall that firms are either honest (never bribe) or strategic, with the former type occurring with probability θ . Firms receive a payoff proportional to the measure of active consumers $0 \leq X_t^* \leq 1$ (i.e., those who purchase the good) less any bribes they pay to the reviewer, b_t . Thus, we can write the total payoff of the time t firm as $X_t^* - b_t$. Firms cannot observe bribes from previous periods but can make inferences about how receptive the reviewer is to bribes based on her reputation $\hat{\omega}$. For example, a reviewer with reputation $\hat{\omega} = H$ is committed to reporting high-quality signals and therefore will not be receptive to bribes. Thus, a (strategic) firm at time t chooses b_t to maximize $\mathbb{E}[X_t^* - b_t | \hat{\omega}]$. Firms also have an outside option $\gamma > 0$ that they give up if they choose to enter the market.

3.4. Equilibrium Characterization

We consider the Bayesian Nash equilibria (BNE) of the aforementioned game, which may or may not be in mixed strategies. Recall that the type H reviewer plays according to the strategy in Section 3.2, which maps signals s_t and bribes b_t to reviews r_t .

In the following result, we characterize the equilibrium strategies of the three agent types.

Theorem 1. All equilibria are in pure strategies and take one of two forms:

(i) *Separating equilibrium:* The reviewer chooses $\hat{\omega} = \omega$, every firm bribes $b_t = 0$, and consumer i purchases if and only if

$$\phi(i) \leq \frac{r_t}{1 + \sigma_\omega^2}.$$

(ii) *Pooling equilibrium:* The reviewer chooses $\hat{\omega} = L$, strategic firms bribe $b_t = b^*$, and consumer i purchases if and only if

$$\phi(i) \leq \frac{r_t}{1 + \sigma_L^2}$$

under generic conditions.⁸ Moreover, the separating equilibrium always exists.

There is either just one equilibrium (separating) or two equilibria (separating and pooling). We call environments

where only the separating equilibrium exists *bribe proof*, because the equilibrium admits no bribes and both reviewer types report their signals truthfully. This is the first-best outcome relative to environments where the pooling equilibrium also exists. With the pooling equilibrium, consumers do not get first-best outcomes because a type H reviewer distorts her signals by playing effective type L.

We decompose the equilibrium strategies of Theorem 1 by agent type.

(i) *Consumers*: Based on the reputation of the reviewer, $\hat{\omega}$, the consumer obtains an estimate of the expected quality given the period t review using the inverse-variance estimator:

$$\mathbb{E}[q_t | r_t, \hat{\omega}] = \frac{r_t / \hat{\sigma}^2}{1 + 1/\hat{\sigma}^2} = \frac{r_t}{1 + \hat{\sigma}^2},$$

where $\hat{\sigma}$ is used as shorthand for $\sigma_{\hat{\omega}}$. The 1 in the denominator is a consequence of assuming $q_t \sim \mathcal{N}(0, 1)$.⁹ The consumer thus tempers the review on both extremes: She believes very good reviews overhype the quality, whereas very negative reviews are overly harsh. Moreover, when the reviewer plays a more-skilled effective type (low $\hat{\sigma}$), all consumers place greater weights on the review and lower weights on their priors.

(ii) *Reviewer*: A type L reviewer plays as her true type in both the separating and pooling equilibria. On the other hand, a type H reviewer has a choice to play either $\hat{\omega} = H$ or $\hat{\omega} = L$. Based on her choice, either the separating or pooling equilibrium survives. This choice is in turn determined by whether the bribe b^* is sufficiently high to compensate for her reduced influence. The reduction in influence makes consumers pay less attention to reviews, which hurts the reviewer in two ways: directly, through the decreased payoffs she gets from the loss in influence, and indirectly, through the reduction in the firms' willingness to bribe for biased reviews.

(iii) *Firms*: Strategic firms observe the reputation $\hat{\omega}$ of the reviewer and decide whether the demand boost they get from a biased review is enough to offer $b_t = b^*$. When $\hat{\omega} = H$, the firms recognize that the reviewer has a strong reputation for honest, high-quality reviews. This makes them abstain from bribing (the separating equilibrium). Conversely, when $\hat{\omega} = L$, the firm weighs the possibility that this is a type H reviewer who accepts bribes and biases reviews (the pooling equilibrium) against the possibility that this is an actual type L reviewer who cannot misreport to help bribing firms (the separating equilibrium).

3.5. Discussion: Commitment Assumption and the Full Dynamic Model

A key assumption of our model is that the reviewer commits upfront (at $t = 0$) to a strategy mapping signals

and bribes to reviews. This is a common assumption when the agent has reputational concerns over how the accuracy of her reviews are perceived.¹⁰ We use the commitment assumption in our model for parsimony and to most transparently demonstrate our main ideas.

That being said, the main findings of Theorem 1 are robust to relaxing the commitment assumption. In Online Appendix B.1, we present a more general dynamic model where the reviewer has full autonomy over how she writes reviews. Although there may be many more equilibria in the fully dynamic setting, we present mild equilibrium refinements in Online Appendix B.2 that lead to the same equilibria of Theorem 1 (as shown in Online Appendices B.3 and B.4). A key finding is that the reviewer must mimic an effective type. This is because consumers *will* learn how the reviewer is playing and will stop listening to her if they realize she is inconsistent with a type L or type H reviewer playing truthfully (see Lemma B.3). Moreover, a low-skilled reviewer ($\omega = L$) must play effective type $\hat{\omega} = L$ because she cannot remove noise from her signal to improve her estimate and make it on par with that of a type H reviewer (Lemma B.7). Thus, similar to the commitment model, it is the type H reviewer who must choose whether to play $\hat{\omega} = H$ or $\hat{\omega} = L$.

Similarly, the assumption that the firm has a binary choice to either **Bribe** ($b_t = b^*$) or **Not Bribe** ($b_t = 0$) instead of choosing a continuous bribe $b_t \in \mathbb{R}_+$ is also without loss of generality. The equilibria of the full dynamic model of Online Appendix B.1, under the refinements of Online Appendix B.2, lead to all firms (who elect to bribe) choosing the same bribe b^* (see Lemmas B.5 and B.6).

Finally, the equivalence of the full dynamic model and the commitment model depends on the assumption of a patient reviewer (i.e., δ close to one). This is a standard assumption in the reputation literature (Fudenberg and Levine 1989), and to retain consistency with this model, we assume $\delta \rightarrow 1$ throughout the remainder of the paper. For the same reason, we look at the long-run outcomes in the commitment model, despite the fact that this is not necessary to derive our results in the commitment setting.

4. How Do Bribes Impact Consumer Utility?

We now examine the effect of bribes on consumer utility. A consumer's decision to purchase in period t is informed by her outside option $\phi(i)$, the review r_t , and the reviewer's reputation $\hat{\omega}$. The review induces a set of active consumers $\mathcal{A}_t$, and consumer i 's utility is nonnegative if she purchases the product whenever

the quality q_i is at least equal to her outside option $\phi(i)$.

Definition 1. Consumer utility at time t is given by the average consumer utility of active consumers conditional on reviews r_t and reputation $\hat{\omega}$, $CU_t = \int_{i \in A_t} (q_i - \phi_i) di$.¹¹ The average consumer utility at time t is the time-average of consumer utility up until time t , $\bar{CU}_t = \frac{1}{t} \sum_{\tau=1}^t CU_\tau$.

The following result establishes that average consumer utility and the reviewer's influence (from Equation (1)) converge to values that only depend on the effective accuracy $\hat{\sigma}$ of the reviewer.

Proposition 1. *The influence of the reviewer and average consumer utility converge (almost surely) as $t \rightarrow \infty$ to $I_\infty(\hat{\sigma})$ and $\bar{CU}_\infty(\hat{\sigma})$, respectively. These quantities only depend on and are strictly decreasing in $\hat{\sigma}$.*

From the previous section, an H type reviewer plays either $\hat{\sigma} = \sigma_H$, which maximizes both her influence and consumer utility, or she plays $\hat{\sigma} = \sigma_L$, in which case both of these quantities are strictly less than their first-best levels.

An H type reviewer therefore weighs the loss of influence from playing $\hat{\omega} = L$ against the bribes she receives from firms. This will determine whether the pooling equilibrium of Theorem 1 survives or if only the separating equilibrium exists. In the latter case, the environment is bribe proof and delivers first-best levels of consumer utility. For the pooling equilibrium, consumer utility is below first-best because information about product quality is obscured by the reviewer.

4.1. Bribe-Proof Environments

We now characterize bribe-proof environments as a function of the parameters of the problem: the proportion θ of honest firms and the information the market has about the comparative skills of the reviewers and the likelihood of the different reviewer types. We study each of these parameters in the next three sections.

4.1.1. Proportion of Honest Firms. We define the *fair consumption* (denoted by $\bar{X}^*$) as the ex ante expected consumption for a firm when the reviewer is of low skill but guarantees a truthful review, that is, $\mathbb{E}_{r_i \sim \mathcal{N}(0, 1 + \sigma_L^2)} \left[\phi^{-1} \left(\frac{r_i}{1 + \sigma_L^2} \right) \right]$. Firms enter the market if the expected consumption of their product is at least equal to their outside option γ . To avoid trivial cases where no firms enter, we assume that the fair consumption of the firm is higher than the outside option.¹² We capture how appealing bribing is to the reviewer in the following definition.

Definition 2. The *bribing propensity*, ψ , is the ratio of the fair consumption (less the outside option) to the

change in influence when a type H reviewer mimics type L, that is, $\psi = (\bar{X}^* - \gamma) / (I_\infty(\sigma_H) - I_\infty(\sigma_L))$.

The bribing propensity ψ serves as a benefit/cost proxy for the reviewer: the numerator $\bar{X}^* - \gamma$ is the amount the strategic firms would gain from bribing (and therefore how much they are willing to pay in bribes) when all firms are strategic and the denominator $I_\infty(\sigma_H) - I_\infty(\sigma_L)$ is how much the reviewer will lose in influence by playing as a lower instead of a higher type. Thus, the higher this ratio is, the more appealing bribing becomes to the reviewer.

When the reviewer has no intrinsic value for influence, that is, $\beta = 0$, then bribes will always exist, because they are the only source of payoff for the reviewer. The next result shows that bribes can still exist even when the reviewer values her influence highly enough. This depends on the proportion of honest firms θ . In particular, the property of being bribe proof is nonmonotone in θ . When most firms are either strategic or honest (i.e., when θ is near zero or one), the environment is bribe proof. However, intermediate values of θ may admit bribes.

Theorem 2. *There exists $0 < \underline{\theta} < \bar{\theta} < 1$ such that:*

- (a) *For all $\beta > \psi$ and $\theta < \underline{\theta}$ or $\theta > \bar{\theta}$, the setting is bribe proof (i.e., only the separating equilibrium exists);*
- (b) *For some $\beta > \psi$, probability of high type p , and bribe b^* , there exists $\underline{\theta} < \theta^* < \bar{\theta}$, such that the pooling equilibrium exists and consumer utility is strictly below first-best levels.*

Theorem 2 describes a phase transition where, as the proportion of honest firms increases, the environment transitions from being bribe proof to being susceptible to bribes and back again. Recall from Section 3.2 that the adjustment noise that the type H reviewer adds (to mimic type L) is drawn from a normal distribution that is parametrized by the proportion of honest firms θ . As θ becomes small enough and less than some $\underline{\theta}$, the bribing firm still receives a mostly fair review, because the reviewer cannot inject much bias when nearly all firms are strategic. This provides the firm with close to fair consumption less the outside option γ . Naturally, a bribe that only provides a fair review (and fair consumption) does not offer much benefit to the firm. Consequently, in equilibrium, all firms offer $b_i = 0$ and the reviewer reports truthfully. Hence, the environment is bribe proof.

At the other end, when $\theta > \bar{\theta}$, the reviewer can inject a lot of positive bias into the reviews of bribing firms. This makes firms willing to offer a bribe. However, because there are only few of these firms, it becomes unprofitable for the reviewer to trade off her influence with the bribes she (infrequently) collects. Instead, she opts to maximize her influence by reporting truthfully. This again leads to a bribe-proof environment.

The intermediate region of θ is where the reviewer and the strategic firms can come to an agreement. The

reviewer incurs the reduced influence from playing type L, and the strategic firms find that the bias they receive is worth offering a bribe. Because θ is in an intermediate region, there are enough strategic firms, and the reviewer collects this bribe fairly often. This makes bribing profitable for both the reviewer and firms. As a result, the reviews misreports and the environment admits bribes.

Figure 2 presents the quintessential cases that can arise. When $\beta > \psi$, Theorem 2(a) applies and we know that either Case I or Case II hold (i.e., the extreme ends of θ are bribe proof). In some cases, as identified in Theorem 2(b) and shown in Case II, there exists an intermediate range of θ where the environment may not be bribe proof although the extremes are. In environments where $\beta < \psi$ (and Theorem 2 does not apply), other situations (e.g., Case III) can occur. For example, the property of being bribe proof can be monotone in the fraction of honest firms θ , although Cases I and II can still occur in these environments.

Theorem 2 offers a warning for policymakers. It suggests that policy interventions that aim to reduce the number of firms that are willing to bribe (e.g., through threats of audits and punishments) can move the whole system from a bribe-proof environment (low θ) to a regime that admits bribes (intermediate θ). This reduces consumer utility and accomplishes the opposite goal of what the intervention is designed for. Similarly, even when the proportion of honest firms is high enough and the environment is bribe proof, a small slip that reduces that proportion of these firms below $\bar{\theta}$ is enough for the environment to revert back to being vulnerable to bribes.

4.1.2. Disparity Between Skill Types. Having established that consumer utility is nonmonotone in the proportion of honest firms, we next show that it is also nonmonotone in the difference of possible reviewer skills. Recall that σ_H and σ_L are the precisions of the high-skilled and low-skilled reviewers, respectively. Assume that the accuracy of the low-skilled reviewer improves (i.e., σ_L decreases) while σ_H remains

the same. Does this make consumers better or worse off? We address this in the following result.

Proposition 2. Let $\Delta = \sigma_L - \sigma_H$. The environment is bribe-proof as $\lim_{\Delta \rightarrow 0}$ or $\lim_{\Delta \rightarrow \infty}$.

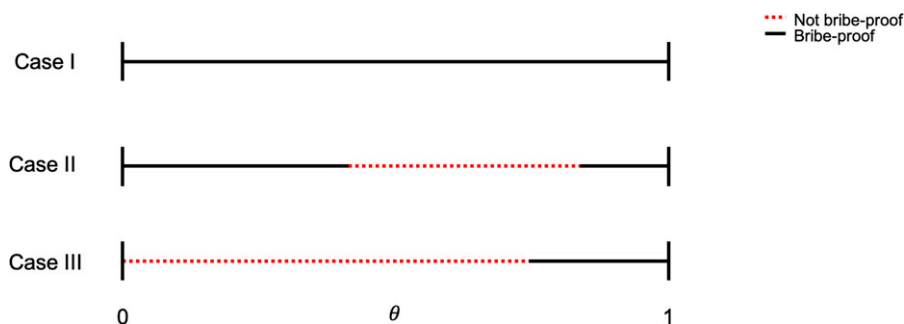
Proposition 2 shows that the answer to this question is not immediately obvious. When the skill of the low type reviewer improves, the environment can transition from being bribe proof to admitting bribes. This occurs because firms do not have enough incentives to bribe when the difference in skills between the reviewer types is too small ($\Delta \rightarrow 0$) or too large ($\Delta \rightarrow \infty$) but may have incentives to do so for intermediate values of Δ . We next describe the details for these three cases.

When the difference is too large (i.e., $\Delta \rightarrow \infty$), the type H reviewer can inject a lot of bias by mimicking a type L. Firms, however, are unwilling to bribe because, although bribing would deliver heavily favorable reviews, consumers pay very little attention to what the reviewer has to say because the type L reviewer being mimicked is extremely inaccurate. Similarly, when σ_H is fixed and $\sigma_L \rightarrow \infty$, the reviewer must sacrifice substantial influence to accept bribes, which requires a more significant bribe from the firm.

Assume that the skill of the type L reviewer increases (i.e., Δ shrinks) and she becomes more accurate. As before, the H type reviewer can still help the bribing firms by mimicking type L and injecting bias into her reviews. The difference with the previous case is that consumers now pay more attention to the reviews because the type L being mimicked has reasonable accuracy. This makes bribing attractive to firms and the environment transitions into admitting bribes.

If the skill of the L type reviewer increases further such that the difference between the types becomes too small (i.e., $\Delta \rightarrow 0$), then the bias an H type reviewer can inject by mimicking an L type is minimal. This is because she is mimicking a skill that is not that different from her own. Firms are not interested in offering a bribe for a review that is too similar to the one they would get without a bribe, and the environment transitions back to being bribe proof.

Figure 2. (Color online) Bribe-Proof Environments as a Function of Proportion of Honest Firms θ



Proposition 2 highlights the fact that the property of being bribe proof is more a matter of relative rather than absolute skill levels. The existence of bribes in the system depends on whether the reviewer can successfully imitate a less informative reviewer without losing her influence over consumer decisions altogether. It establishes that improvement in skill levels—which would normally improve the accuracy of information about quality—need not improve consumer utility. In fact, this skill improvement can lead to more information distortion when bribes exist in equilibrium.

4.1.3. Prior Beliefs About Skill. We conclude our discussion of bribe-proof environments by discussing how the incentives of players respond to the likelihood of the reviewer being highly skilled. In this section, we show that a reviewer who is more likely to be low skilled can be better for consumers than a reviewer who is more likely to be high skilled.

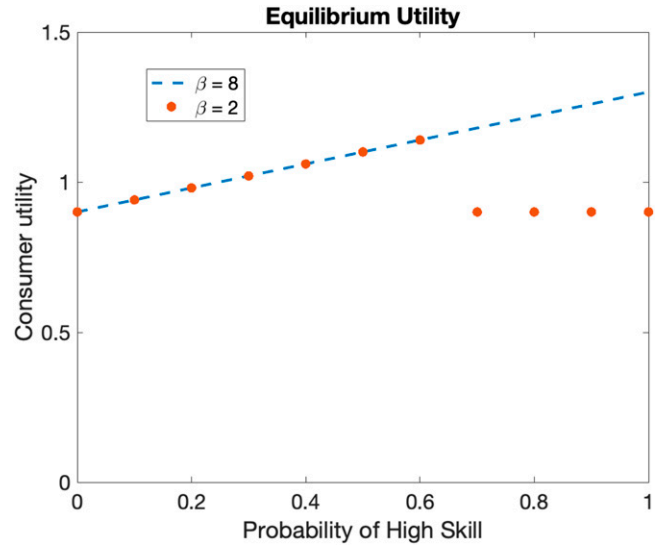
Recall that the probability p indicates the likelihood that the true type of the reviewer is H. We consider what happens to consumer utility as p increases. When there are no bribing firms (i.e., $\theta = 1$), increasing p always improves consumer utility. Because all firms are honest, reviewers report truthfully (i.e., the separating equilibrium is unique), and consumers get more accurate recommendations. However, with the introduction of firms that might bribe, an increase in the likelihood of the type H reviewer also increases the potential for bribes and information distortion. This is because an actual type H reviewer is able to inflate the reviews of the bribing firms, whereas an actual type L cannot. Thus, as p increases, the incentives for firms to bribe become stronger and the environment may no longer be bribe proof. This is summarized in the following result.

Proposition 3 (Prior Beliefs). *There exists p^* such that when $p < p^*$ the environment is bribe proof (unique separating equilibrium) and consumer utility is increasing in p . However, $p > p^*$ consumer utility is minimized (pooling equilibrium exists) and strictly below the first-best levels when $p < p^*$.*

Proposition 3 shows that an increase in the prior probability p of a high type reviewer introduces two competing effects: it increases the potential for more precise reviewers, but it also increases the incentives of the firm to bribe and corrupt the reviews. Thus, although increasing p improves the quality of the information the reviewer receives (on average), it also encourages bribery and therefore can reverse the anticipated improvement in consumer utility.

The property that increasing the expected skills of the reviewer can lead to a reduction in (average)

Figure 3. (Color online) Equilibrium Utility for Varying Ex Ante Skill



consumer utility can be seen in Figure 3. In this figure, p varies from zero to one for different values of the influence propensity β of the reviewer. When $p = 0$, the reviewer is always low skill, and no bribes are offered because firms have nothing to gain from bribing. When p is close to one, the firm knows that the reviewer is likely a type H reviewer. Observing this reviewer playing $\hat{\omega} = L$ makes firms almost certain that this is a reviewer who is biasing reviews in exchange for bribes, which makes firms more likely to offer them.

Whether the environment is bribe proof is determined by the value of β . When β is large (dashed line in Figure 3), the reviewer places a very high value on her influence and cannot be swayed by bribes. In this case, increasing p (and the expected accuracy of the reviewer) can only increase consumer utility. This is because consumers receive better information about product qualities and this information is never distorted.

However, when β is not too large (dotted line), there is a more subtle effect from increasing p . As p increases, firms are more willing to bribe knowing there is a higher likelihood that the reviewer is a true type H playing as effective type L and biasing her reviews. Eventually (around $p = 0.6$ in the figure), the firm is willing to pay b^* as a bribe, and with smaller β , the bribe b^* is enough to entice the reviewer to “sell out” and choose $\hat{\omega} = L$. Before that point, increasing p increases the accuracy of reviews (as in the case of high β) and increases consumer utility. However, once p hits the threshold, the firm becomes willing to bribe knowing the reviewer likely provides biased reviews in exchange for bribes ($\hat{\omega} = L$ even though $\theta = H$, in the pooling

equilibrium). This causes a precipitous drop to the lowest consumer utility at this threshold, as both skill types play identically as the low type in equilibrium.

Figure 3 illuminates an important counterintuitive finding: improving the quality of information and efficacy of reviews may not necessarily lead to better outcomes for consumers. This is because firms can also leverage this improvement for their own gain, and together with reviewers, extract greater surplus through bribes at the cost of hurting consumers.

4.2. Total Welfare Analysis

Until now, we used consumer welfare as our metric for comparing good versus bad outcomes. In certain circumstances, it might be prudent to look at the *total* welfare of all market participants, that is, the sum of utilities of consumers, reviewers, and firms. The net utility of the reviewer is $\beta \cdot I_t$ in addition to any bribes that she may receive. Likewise, the net utility of the firm is the consumption X_t^* less any amount that is paid out to the reviewer in bribes. Thus, the bribes cancel out, and in addition to the consumer utility CU_t studied until now, we also include the payoffs for the reviewers and the firms. Formally, the total welfare is given by

$$W \equiv \frac{1}{t} \sum_{\tau=1}^{\infty} \left[\underbrace{\beta \cdot I_{\tau} + b_{\tau}}_{\text{Reviewer}} + \underbrace{X_{\tau}^* - b_{\tau}}_{\text{Firm}} + \underbrace{CU_{\tau}}_{\text{Consumer}} \right].$$

Given this, we obtain a generalization of the welfare analysis presented.

Proposition 4. *If ϕ is symmetric around the average consumer at $1/2$ (i.e., $\phi(1/2 + z) = -\phi(1/2 - z)$), then an equilibrium with bribes (i.e., pooling) has strictly lower welfare for firms, consumers, and in total than an equilibrium with no bribes (i.e., separating).*

In other words, Proposition 4 states that under mild conditions on ϕ , bribe proof is best not only for consumers and the market as a whole but for *the firms themselves*. This means that firms, on average, are hurt by the existence of bribes in the system. The reason is that when ϕ is symmetric, total firm consumption across both honest and strategic firms remains unaffected when the reviewer increases the variance of her reviews. Consumers simply temper their interpretation of the review in their purchasing decision, and there is no net effect in the expected consumption over time. In a bribe-proof environment, firm welfare is given by consumption of its product. When there are bribes, firm welfare is given by consumption minus the bribe they have to pay. Because bribes are positive and average consumption (across all firms) remains the same, firm welfare decreases.

It is clear that honest firms are hurt when the environment is not bribe proof, because their reviews are

consistently biased down. Interestingly, the *bribing* firms themselves could also be hurt and would prefer if bribes were eliminated from the system altogether. To see why this is the case, consider the following two settings where half the firms are honest and the other half are strategic (similar to the setting of Section 2):

- *Setting A:* This is the typical setting where firms can bribe in exchange for biased reviews (as in the pooling equilibrium of Theorem 1). A firm that bribes receives a review that is biased up (distributed as the top-left figure of Figure 1), whereas a firm that does not bribe receives a review that is biased down (distributed as the top-right figure of Figure 1). The difference in expected consumption from these two reviews determines whether a strategic firm will pay b^* to receive a biased review.

- *Setting B:* For the sake of demonstration, suppose all firms agree upfront not to bribe and that this agreement is enforceable. Firms receive an honest review from the distribution given in the bottom figure of Figure 1. This review provides a tighter estimate of the quality q_t for consumers, as the reviewer commits to $\hat{\omega} = H$ instead of injecting bias. Moreover, the firm does not pay a bribe to receive this review.

In Setting A, the firm may bribe simply to avoid a bad review. It may be the case that a fair review plus not having to pay a bribe (i.e., Setting B) is more profitable than the biased review that the firm solicits with bribe b^* (Setting A). However, the threat of the bad review is enough to force the strategic firms to bribe in equilibrium. In these cases, the bribe that the firm pays to the reviewer is a form of extortion: it offers no benefit to the firm but rather saves it from being punished with a bad review. As a result, eliminating bribes can be a Pareto improvement for *all* firms in the market.

5. Final Remarks

Review manipulation is a problem that takes on multiple forms. In this paper, we focused on the case where an influential reviewer can misrepresent her reviews to favor specific firms. This decreases the utility of consumers as they purchase products of inferior quality and avoid products of high quality. The reviewer achieves this through a novel mechanism that takes advantage of the statistical learning process of consumers. This mechanism allows us to derive insights into the long-run nature of these interactions. For example, we show that an increase in the number of firms that are not willing to bribe (e.g., through a stringent auditing policy) might make the market more vulnerable to bribes and review distortion. We also show that improving the quality of the reviewer can result in worse outcomes for consumers.

An important factor that prevents bribes in our model is that the reviewer can generate payoffs from

consumers via her influence (captured by the parameter β). This is a channel that is becoming more common in practice, as many online reviewers receive recurring payments directly from consumers through platforms like Patreon. Thus, a combination of low auditing and higher “wages” can improve welfare. This conclusion is similar to the literature on corruption, as in the work of Becker and Stigler (1974) and the empirical documentation in Di Tella and Scharrotsky (2003).

Our model can be extended in several directions. In Online Appendix B, we show that our results hold even when the reviewer does not commit to playing a specific strategy but can instead deploy any dynamic strategy she chooses. Under mild assumptions, we show that the reviewer must still play using the same commitment strategy as in the model in this paper, and all the insights from that model continue to hold.

Another extension is competition between reviewers. Our setup considered a monopolist reviewer. When there are multiple reviewers, it can be shown that a firm will not bribe more than a single reviewer in the same period. This is because the long-run dynamics will eventually show that biased reviews are correlated over time. Consequently, this will reveal which reviewers are accepting bribes and which firms are offering them. An unexpected consequence of competition is that a market with a monopolist reviewer may be bribe proof but starts admitting bribes when another reviewer is added. This is because the new reviewer “steals” some influence from the monopolist reviewer, who now starts accepting bribes to make up for this decreased influence.¹³

Finally, we remark that our work is applicable beyond the setup of online reviews. Thinktanks and research institutions that obtain funding from outside sources face similar tradeoffs: they want to produce reliable results for the public, but they risk cutting off their funding if these results do not advance the interests of the funding sources. Another area of application is the inspection and certification market, which is projected to be worth more than \$400 billion by 2025 (Bloomberg 2019). These markets cover a wide range of industries (manufacturing, agriculture, healthcare, etc.) and can be modeled using the framework developed in this paper. Despite the importance of these interactions and their impact on all parties involved, their dynamic nature have not been analyzed or perfectly understood. We hope that the model in this paper serves as a first step in this direction.

Appendix A. Proofs

Throughout the proofs, we will use the following shorthand notation: Let $\mathcal{N}(0, \sigma^2, \underline{z}, \bar{z})$ denote the conditional normal distribution with a mean of zero and a variance of σ^2

but where the value is drawn between the $\underline{z}$ and $\bar{z}$ percentiles of the distribution.

A.1. Main Body

Proof of Theorem 1. Because the reviewer commits to and announces her effective type $\hat{\omega}$, her strategy of mapping signals and bribes to reviews is fixed. This means that the game is stationary. Thus, it is sufficient to consider one period in the equilibrium analysis.

The roadmap for the proof is as follows. First, we derive the consumers’ best response, which depends only on the reviewer’s choice of effective type $\hat{\omega}$ (her “effective accuracy”). Second, we show that the a high type reviewer must give up influence when she elects $\hat{\omega} = L$ instead of $\hat{\omega} = H$. Of the pure strategy equilibria, we call the ones where $\hat{\omega} = \omega$ the *separating* case, and the ones with $\hat{\omega} = L$ for both types the *pooling* case. Third, we show that of the two possible separating cases (corresponding to **Not Bribe** and **Bribe**), only the **Not Bribe** case can be supported in equilibrium. Of the two possible pooling cases, only **Bribe** can be supported. Fourth, we show that generically, no mixed-strategy equilibria can exist, so only the unique separating and unique pooling pure-strategy equilibria can exist. We conclude by proving that the separating equilibrium is in fact *always* an equilibrium (whereas the pooling one may or may not be).

Because consumers cannot observe the bribes and are myopic, they simply purchase the good if $\phi(i) \leq \mathbb{E}[q_t | r_t, \hat{\omega}]$. We note that consumers can compute $\mathbb{P}[q_t | r_t]$ (where $\mathbb{P}[\cdot]$ represents the density) using Bayes’ rule given that the reviewer writes $r_t = q_t + \varepsilon_t$ where $\varepsilon_t \sim \mathcal{N}(0, \hat{\sigma}^2)$ ¹⁴:

$$\begin{aligned} \mathbb{P}[q_t | r_t] &= \frac{\mathbb{P}[r_t | q_t] \mathbb{P}[q_t]}{\int_{-\infty}^{\infty} \mathbb{P}[r_t | q] \mathbb{P}[q] dq} \\ &= \frac{\exp\left(-\frac{(r_t - q_t)^2}{2\hat{\sigma}^2}\right) \exp\left(-\frac{q_t^2}{2}\right)}{\int_{-\infty}^{\infty} \exp\left(-\frac{(r_t - q)^2}{2\hat{\sigma}^2}\right) \exp\left(-\frac{q^2}{2}\right) dq} \\ &= \frac{1}{\sqrt{2\pi\hat{\sigma}}} \frac{\exp\left(-\frac{(r_t - q_t)^2}{2\hat{\sigma}^2}\right) \exp\left(-\frac{q_t^2}{2}\right)}{\exp\left(-\frac{r_t^2}{2(\hat{\sigma}^2 + 1)}\right) / \sqrt{\hat{\sigma}^2 + 1}} \\ &= \frac{1}{\sqrt{2\pi}} \sqrt{\frac{\hat{\sigma}^2 + 1}{\hat{\sigma}^2}} \exp\left(-\frac{\left(q_t - \frac{r_t}{1 + \hat{\sigma}^2}\right)^2}{2\hat{\sigma}^2/(1 + \hat{\sigma}^2)}\right), \end{aligned}$$

which is the normal distribution density function with mean $\frac{r_t}{1 + \hat{\sigma}^2}$ and variance $\frac{\hat{\sigma}^2}{1 + \hat{\sigma}^2}$. Thus, $\mathbb{E}[q_t | r_t, \hat{\omega}] = \frac{r_t}{1 + \hat{\sigma}^2}$, identifying the consumer’s strategy stated in Theorem 1.

We now show that a type H reviewer loses some influence when she plays $\hat{\omega} = L$. Let $\hat{\sigma}$ be the effective precision of the reviewer. Given the consumer’s strategy we had just identified and recalling that it depends on $\mathbb{E}[q_t | r_t, \hat{\omega}]$, we can write the reviewer’s expected influence as

$$\mathbb{E}_{r_t}[I_t] = \frac{1}{\sqrt{2\pi(1 + \hat{\sigma}^2)}} \int_{-\infty}^{\infty} \left(\phi^{-1}\left(\frac{r_t}{1 + \hat{\sigma}^2}\right) - \phi^{-1}(0) \right)^2 \cdot \exp\left(-\frac{r_t^2}{2(1 + \hat{\sigma}^2)}\right) dr_t.$$

Let us make the substitution $\alpha = r_t / \sqrt{1 + \hat{\sigma}^2}$:

$$\mathbb{E}_{r_t}[I_t] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\phi^{-1}\left(\frac{\alpha}{\sqrt{1 + \hat{\sigma}^2}}\right) - \phi^{-1}(0) \right)^2 \cdot \exp(-\alpha^2/2) d\alpha \quad (\text{A.1})$$

Note that $\phi^{-1}(\cdot)$ is an increasing function, and $|\alpha / \sqrt{1 + \hat{\sigma}^2}|$ is decreasing in $\hat{\sigma}$, so the integrand is decreasing in $\hat{\sigma}$ pointwise for all α . Thus, worse precision $\hat{\sigma}$ (i.e., higher $\hat{\sigma}$) leads to lower influence.

We use this result to analyze the game between the reviewer and firms. The type L reviewer has no choice in her strategy and the type H reviewer has two choices. Thus, there are two potential (pure-strategy) maps from reviewer types ω to strategies $\hat{\omega}$. We call the map where $\hat{\omega} = \omega$ the *separating* case and $\hat{\omega} = L$ for all ω the *pooling* case. In the former, all reviewer types report truthfully and do not differentiate based on bribes, so **Not Bribe** is a best response for the firm. In the latter, all reviewer types play according to L . If the firm were to play **Not Bribe** with a probability of one, this would not be an equilibrium: because $\beta > 0$ (generic conditions rule out $\beta = 0$), the type H reviewer gives up influence when choosing $\hat{\omega} = L$ but does not get compensated with any bribes. Therefore, there is no equilibrium where the firms choose **Not Bribe** with a probability of one, and the type H reviewer plays $\hat{\omega} = L$. Thus, the pure pooling equilibrium is the following: the firm chooses **Bribe**, both types play $\hat{\omega} = L$, and the consumer buys if $\phi(i) \leq r_t / (1 + \sigma_L^2)$.

Next, we show that generically, no mixed strategy equilibria exist. If the firm totally mixes between **Bribe** and **Not Bribe**,¹⁵ it must be indifferent between **Bribe** and **Not Bribe**; in other words,

$$\begin{aligned} & \mathbb{E}_{q_t, \varepsilon_t, \varepsilon'_t \sim \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, \theta)} \left[\phi^{-1}\left(\frac{r_t}{1 + \hat{\sigma}^2}\right) \right] \\ &= \mathbb{E}_{q_t, \varepsilon_t, \varepsilon'_t \sim \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, \theta, 1)} \left[\phi^{-1}\left(\frac{r_t}{1 + \hat{\sigma}^2}\right) \right] - b^*, \end{aligned}$$

which occurs on a zero measure set. Therefore, generically, in the pooling case, the firm must choose **Bribe** with a probability of one.

Next, we show that the type H reviewer does not mix between $\hat{\omega} = H$ and $\hat{\omega} = L$ either. Suppose the $\omega = H$ reviewer plays $\hat{\omega} = H$ with probability $\zeta \in (0, 1)$ and $\hat{\omega} = L$ with probability $1 - \zeta$. From the same reasoning as before, when the reviewer plays $\hat{\omega} = L$, it must be the case that firms play **Bribe** with a probability of one. If the reviewer is randomizing, then her expected payoff from playing $\hat{\omega} = L$ and $\hat{\omega} = H$ should be equal:

$$\begin{aligned} & \mathbb{E}_{r_t} \left[(1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} \beta I_t + b_t \mid \hat{\omega} = L \right] = \mathbb{E}_{r_t} \left[(1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} \beta I_t + b_t \mid \hat{\omega} = H \right] \\ & \Rightarrow \beta \mathbb{E}_{r_t}[I_t \mid \hat{\omega} = L] + b^* = \beta \mathbb{E}_{r_t}[I_t \mid \hat{\omega} = H], \end{aligned}$$

where the expressions $\mathbb{E}_{r_t}[I_t \mid \hat{\omega} = L]$ and $\mathbb{E}_{r_t}[I_t \mid \hat{\omega} = H]$ are time independent because of the stationarity of the game (and given by the expression in Equation (2)). This equality holds only on a set of measure zero, so the reviewer has a strict preference for either $\hat{\omega} = H$ or $\hat{\omega} = L$, and only

the two pure-strategy equilibria (separating and pooling) can exist.

Finally, we show that the separating case is always an equilibrium. Suppose that firms believe $\hat{\omega} = \omega$ is played in equilibrium. When the firm observes $\hat{\omega} = H$, it knows that this is a true type H reviewer who reports truthfully. When the firm observes $\hat{\omega} = L$, it knows it is a true type L reviewer who reports truthfully. In either case, it is a best response for the firm to select **Not Bribe**. Given the firm picks **Not Bribe**, the true type H reviewer who selects $\hat{\omega} = L$ gives up influence but does not receive any bribes. This is not a profitable deviation, and hence, the separating case is always an equilibrium. $\square$

Proof of Proposition 1. Because $\mathbb{E}[q_\tau \mid r_\tau, \hat{\omega}]$ are i.i.d. random variables for the consumers, we know by the strong law of large numbers that

$$\begin{aligned} & \frac{1}{t} \sum_{\tau=1}^t (\phi^{-1}(\mathbb{E}[q_\tau \mid r_\tau, \hat{\omega}]) - \phi^{-1}(0))^2 \\ & \xrightarrow{a.s.} \mathbb{E}_{q_\tau, r_\tau} [(\phi^{-1}(\mathbb{E}[q_\tau \mid r_\tau, \hat{\omega}]) - \phi^{-1}(0))^2] = I_\infty(\hat{\sigma}), \end{aligned}$$

where the final equality can be seen by simply integrating out q_τ and noting that $I_\infty(\hat{\sigma})$ for effective type $\hat{\omega}$ playing with precision $\hat{\sigma}$ is given exactly by $\mathbb{E}_{r_t}[I_t \mid \hat{\omega}]$ from Theorem 1. Thus, this implies $I_\infty(\hat{\sigma})$ is decreasing in $\hat{\sigma}$.

Similarly, for consumer utility, we can write

$$\bar{C}U_t = \frac{1}{t} \sum_{\tau=1}^t CU_\tau = \frac{1}{t} \sum_{\tau=1}^t \int_0^1 (q_\tau - \phi(i)) \cdot \mathbf{1}_{\mathbb{E}[q_\tau \mid r_\tau, \hat{\omega}] \geq \phi(i)} di,$$

which are i.i.d. random variables, so by the strong law of large numbers,

$$\bar{C}U_t \xrightarrow{a.s.} \int_0^1 \mathbb{E}_{q_\tau, r_\tau} [(q_\tau - \phi(i)) \cdot \mathbf{1}_{\mathbb{E}[q_\tau \mid r_\tau, \hat{\omega}] \geq \phi(i)}] di \equiv \bar{C}U_\infty(\hat{\sigma}).$$

Any consumer i with outside option $\phi(i)$ is an active consumer at time t if and only if $\mathbb{E}[q_t \mid r_t, \hat{\omega}] = r_t / (1 + \hat{\sigma}^2) \geq \phi(i)$, or in other words, $r_t \geq (1 + \hat{\sigma}^2)\phi(i)$. The average utility as $t \rightarrow \infty$ can be measured as

$$\bar{C}U_i = \frac{1}{2\delta\pi} \int_{-\infty}^{\infty} \int_{(1+\hat{\sigma}^2)\phi(i)}^{\infty} (q - \phi(i)) \exp(-q^2/2) \exp\left(-\frac{(r-q)^2}{2\hat{\sigma}^2}\right) dr dq.$$

By Fubini's theorem, let us reverse the order of integration and evaluate

$$\begin{aligned} \bar{C}U_i &= \frac{1}{2\delta\pi} \int_{(1+\hat{\sigma}^2)\phi(i)}^{\infty} \int_{-\infty}^{\infty} (q - \phi(i)) \exp(-q^2/2) \exp\left(-\frac{(r-q)^2}{2\hat{\sigma}^2}\right) dq dr \\ &= \frac{1}{\sqrt{2\pi}(1+\hat{\sigma}^2)^{3/2}} \int_{(1+\hat{\sigma}^2)\phi(i)}^{\infty} (r - (1+\hat{\sigma}^2)\phi(i)) \exp\left(-\frac{r^2}{2(1+\hat{\sigma}^2)}\right) dr. \end{aligned}$$

Let us make the same change of variables $\alpha = r / \sqrt{1 + \hat{\sigma}^2}$:

$$\begin{aligned} \bar{C}U_i &= \frac{\sqrt{1 + \hat{\sigma}^2}}{\sqrt{2\pi}(1 + \hat{\sigma}^2)^{3/2}} \int_{\sqrt{1 + \hat{\sigma}^2}\phi(i)}^{\infty} \left(\sqrt{1 + \hat{\sigma}^2}\alpha - (1 + \hat{\sigma}^2)\phi(i) \right) \\ & \quad \times \exp(-\alpha^2/2) d\alpha \\ &= \frac{1}{\sqrt{2\pi}(1 + \hat{\sigma}^2)} \int_{\sqrt{1 + \hat{\sigma}^2}\phi(i)}^{\infty} \left(\alpha - \sqrt{1 + \hat{\sigma}^2}\phi(i) \right) \exp(-\alpha^2/2) d\alpha. \end{aligned}$$

Making a final change of variables to $\kappa = \alpha - \sqrt{1 + \hat{\sigma}^2} \phi(i)$, we have that

$$\begin{aligned}\bar{C}U_i &= \frac{1}{\sqrt{2\pi(1 + \hat{\sigma}^2)}} \int_0^\infty \kappa \exp\left(-\frac{(\kappa + \sqrt{1 + \hat{\sigma}^2} \phi(i))^2}{2}\right) d\kappa \\ &= \frac{1}{\sqrt{2\pi(\hat{\sigma}^2 + 1)}} \exp\left(-\frac{\phi(i)^2(1 + \hat{\sigma}^2)}{2}\right) - \frac{1}{2} \phi(i) \operatorname{erfc}\left(\frac{\phi(i)\sqrt{1 + \hat{\sigma}^2}}{\sqrt{2}}\right),\end{aligned}$$

where $\operatorname{erfc}(\cdot)$ is the complementary error function, $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-x^2} dx$. Let us write $\zeta = \sqrt{1 + \hat{\sigma}^2}$, so that

$$\bar{C}U_i = \frac{1}{\sqrt{2\pi}\zeta} \exp\left(-\frac{\phi(i)^2\zeta^2}{2}\right) - \frac{1}{2} \phi(i) \operatorname{erfc}\left(\frac{\phi(i)\zeta}{\sqrt{2}}\right).$$

Differentiating with respect to ζ , we obtain

$$[\partial\zeta] : -\left(\sqrt{\frac{2}{\pi}} + \frac{1}{\sqrt{2\pi}\zeta^2}\right) \exp\left(-\frac{\phi(i)^2\zeta^2}{2}\right) < 0.$$

Therefore, $\bar{C}U_i$ is decreasing in ζ for all i , which implies it is decreasing in $\hat{\sigma}$. Hence, the average consumer utility $\bar{C}U_\infty$ is also decreasing in $\hat{\sigma}$. $\square$

Proof of Theorem 2. We present the proofs of part (a) and part (b) separately:

(a) The firm's bribe is bounded above by $1 - \gamma$, so the expected bribe (i.e., $(1 - \theta)b^*$) received by the reviewer is at most $(1 - \theta) \cdot (1 - \gamma)$ in any pooling equilibrium. Because $\beta > 0$ and $I_\infty(\sigma_H) > I_\infty(\sigma_L)$ (see Proposition 1), it is clear that for $(1 - \theta) < (1 - \bar{\theta}) \equiv \frac{1 - \gamma}{\beta(I_\infty(\sigma_H) - I_\infty(\sigma_L))}$, the high type reviewer would prefer being truthful to mimicking $\hat{\omega} = L$ and receiving bribes (which holds for $\theta > \bar{\theta}$).

Similarly, when $\theta \rightarrow 0$, if the reviewer is high-skill mimicking low-skill, we see that the firm receives (net) expected consumption from bribing equal to $\bar{X}^* - \gamma$, because by Theorem 1, the firm receives review $r_t \sim q_t + \varepsilon_t + \varepsilon'_t$, where $\varepsilon'_t \sim \lim_{\theta \rightarrow 0} \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, \theta, 1) = \mathcal{N}(0, \sigma_L^2 - \sigma_H^2)$, which implies that $r_t \sim \mathcal{N}(0, 1 + \sigma_L^2)$ and total expected consumption is given simply by $\bar{X}^*$. Conversely, if the firm does not bribe, it receives $r_t \sim q_t + \varepsilon_t + \varepsilon'_t$, where $\varepsilon'_t \sim \lim_{\theta \rightarrow 0} \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, 0, \theta) = \operatorname{Dirac}(-\infty)$, so total expected consumption is zero, which falls below the firm's outside option γ . Thus, the firm does not enter the market and receives a payoff of the outside option γ . The difference in these gives the maximal bribe of $\bar{X}^* - \gamma$. If $b^* > \bar{X}^* - \gamma$, the environment is trivially bribe proof, so assume that $b^* < \bar{X}^* - \gamma$. When the reviewer supplies bias in exchange for bribes, she loses an (average) influence payoff of

$$\begin{aligned}\beta(I_\infty(\sigma_H) - I_\infty(\sigma_L)) &> \psi(I_\infty(\sigma_H) - I_\infty(\sigma_L)), \\ &= \bar{X}^* - \gamma.\end{aligned}$$

As $\theta \rightarrow 1$ the environment is bribe proof because the loss in influence payoff (strictly) exceeds the $\bar{X}^* - \gamma > (1 - \theta)b^*$ for all θ . By continuity we can find a lower bound on θ , such that the environment is bribe-proof for all $\theta < \underline{\theta}$.

(b) We will find θ^* for which the environment is not bribe proof (i.e., a pooling equilibrium exists). For this, we find some β, p, b^* where this pooling equilibrium exists. It is enough to find β and p such that the highest bribe the

“average” firm (i.e., $(1 - \theta)b_t$ for bribing firms offering b_t) is willing to pay is lower than lowest bribe the reviewer is willing to accept (as then there exists some b^* between where a pooling equilibrium does exist).

For ease of notation, we call $v \equiv \frac{1}{2\pi\sqrt{(1 + \sigma_H^2)(\sigma_H^2 - \sigma_L^2)}}$. When the proportion of truthful firms is θ , the total expected consumption obtained from bribing is given by

$$\begin{aligned}\frac{1}{(1 - \theta)v} \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{\phi^{-1}\left(\frac{s + \varepsilon'}{1 + \sigma_L^2}\right)}{\sqrt{\sigma_H^2 - \sigma_L^2} \cdot \Phi^{-1}(\theta)} \cdot \exp\left(-\frac{s^2}{2(1 + \sigma_H^2)}\right) \\ \cdot \exp\left(-\frac{(\varepsilon')^2}{2(\sigma_L^2 - \sigma_H^2)}\right) d\varepsilon' ds,\end{aligned}$$

which we denote as $\kappa(\theta)$. Conversely, the expected consumption from not bribing is given by

$$\begin{aligned}\frac{1}{\theta v} \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{\phi^{-1}\left(\frac{s + \varepsilon'}{1 + \sigma_L^2}\right)}{\sqrt{\sigma_H^2 - \sigma_L^2} \cdot \Phi^{-1}(\theta)} \cdot \exp\left(-\frac{s^2}{2(1 + \sigma_H^2)}\right) \\ \cdot \exp\left(-\frac{(\varepsilon')^2}{2(\sigma_L^2 - \sigma_H^2)}\right) d\varepsilon' ds,\end{aligned}$$

which we denote by $\zeta(\theta)/\theta$. The firm is willing to pay up to the bribe $(p\kappa(\theta) + \min\{-p\zeta(\theta)/\theta, (1 - p)\bar{X}^* - \gamma\})$, which means the average bribe from firms is $\bar{B}(\theta) \equiv (1 - \theta)(p\kappa(\theta) + \min\{-p\zeta(\theta)/\theta, (1 - p)\bar{X}^* - \gamma\})$. Differentiating with respect to θ and applying the fundamental theorem of calculus:

$$\begin{aligned}\frac{\partial \bar{B}^*(\theta)}{\partial \theta} &= -p\eta(\theta) + p\left[\frac{1}{\theta^2}\zeta(\theta) - \frac{1 - \theta}{\theta}\eta(\theta)\right] \cdot \mathbf{1}_{\zeta(\theta)/\theta \geq \gamma/p - (1 - p)/p\bar{X}^*} \\ &\quad + (\gamma - (1 - p)\bar{X}^*) \mathbf{1}_{\zeta(\theta)/\theta < \gamma/p - (1 - p)/p\bar{X}^*} \\ &= \frac{p}{\theta} \cdot (\zeta(\theta)/\theta - \eta(\theta)) \cdot \mathbf{1}_{\zeta(\theta)/\theta \geq \gamma/p - (1 - p)/p\bar{X}^*} \\ &\quad + (\gamma - (1 - p)\bar{X}^* - \eta(\theta)) \cdot \mathbf{1}_{\zeta(\theta)/\theta < \gamma/p - (1 - p)/p\bar{X}^*},\end{aligned}$$

where

$$\begin{aligned}\eta(\theta) &\equiv \frac{\partial \Phi^{-1}(\theta)/\partial \theta}{2\pi\sqrt{1 + \sigma_H^2}} \cdot \exp\left(-(\Phi^{-1}(\theta))^2/2\right) \\ &\quad \cdot \int_{-\infty}^\infty \phi^{-1}\left(\frac{s + \sqrt{\sigma_H^2 - \sigma_L^2} \cdot \Phi^{-1}(\theta)}{1 + \sigma_L^2}\right) \cdot \exp\left(-\frac{s^2}{2(1 + \sigma_H^2)}\right) ds,\end{aligned}$$

and noting that $[(1 - \theta)\kappa(\theta)]' = -\eta(\theta)$ and $\zeta'(\theta) = \eta(\theta)$. By the inverse function theorem, one can see that $\partial \Phi^{-1}(\theta)/\partial \theta = \sqrt{2\pi} \exp((\Phi^{-1}(\theta))^2/2)$, so this reduces to

$$\begin{aligned}\eta(\theta) &= \frac{1}{\sqrt{2\pi(1 + \sigma_H^2)}} \cdot \int_{-\infty}^\infty \phi^{-1}\left(\frac{s + \sqrt{\sigma_H^2 - \sigma_L^2} \cdot \Phi^{-1}(\theta)}{1 + \sigma_L^2}\right) \\ &\quad \cdot \exp\left(-\frac{s^2}{2(1 + \sigma_H^2)}\right) ds.\end{aligned}$$

Because $\lim_{x \rightarrow -\infty} \mathbb{E}[\phi^{-1}(y) | y \leq x] = 0$, we know that $\lim_{\theta \rightarrow 0} \zeta(\theta)/\theta = 0$. Similarly because $\lim_{\theta \rightarrow 0} \Phi^{-1}(\theta) = -\infty$, we see $\lim_{\theta \rightarrow 0} \eta(\theta) = 0$. Therefore, fixing some small $\gamma > 0$, we obtain on an open interval $\theta \in (0, \underline{\theta})$, where $\partial \bar{B}(\theta)/\partial \theta > 0$ for p sufficiently close to one (as then $\zeta(\theta)/\theta < \gamma/p$

$-(1-p)/p \cdot \bar{X}^*$). This implies there exists $\theta^* \in (0, \underline{\theta})$ such that $\bar{B}(\theta^*) > \lim_{\theta \rightarrow 0} \bar{B}(\theta)$. Choosing β such that

$$\psi = \frac{\bar{X}^* - \gamma}{I_\infty(\sigma_H) - I_\infty(\sigma_L)} \equiv \frac{\lim_{\theta \rightarrow 0} \bar{B}(\theta)}{I_\infty(\sigma_H) - I_\infty(\sigma_L)} < \beta < \frac{\bar{B}(\theta^*)}{I_\infty(\sigma_H) - I_\infty(\sigma_L)}$$

implies the reviewer receives a large enough bribe under θ^* to compensate her for her loss in influence. Thus, the environment with θ^* is not bribe proof. $\square$

Proof of Proposition 2. Recall the adjustment noise the reviewer injects is given by $\varepsilon'_t = \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, 0, \theta)$ when she does not get a bribe and $\varepsilon'_t = \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, \theta, 1)$ when she does. Observe that both converge to a Dirac-delta function at zero as $\Delta \rightarrow 0$ (i.e., as $\sigma_H \rightarrow \sigma_L$). In particular, they converge to the same distribution, and thus $\mathbb{E}[X'_t]$ converges to the fair consumption $\bar{X}^*$ regardless of the firm's action of **Bribe** or **Not Bribe**. Hence, for any fixed $b^* > 0$, for σ_H sufficiently close to σ_L , the firm has a best response to choose **Not Bribe**.

When the reviewer mimics low type, we know the payoff from a bribe is given by

$$\mathbb{E}_{q_t, \varepsilon_t, \varepsilon'_t \sim \mathcal{N}(0, \sigma_H^2 - \sigma_L^2, \theta, 1)} \left[\phi^{-1} \left(\frac{q_t + \varepsilon_t + \varepsilon'_t}{1 + \sigma_L^2} \right) \right] - \mathbb{E}_{q_t, \varepsilon_t, \varepsilon'_t \sim \mathcal{N}(0, \sigma_H^2 - \sigma_L^2, 0, \theta)} \left[\phi^{-1} \left(\frac{q_t + \varepsilon_t + \varepsilon'_t}{1 + \sigma_L^2} \right) \right].$$

If $\Delta \rightarrow \infty$, then it must be that $\sigma_L \rightarrow \infty$, and by Lebesgue's dominated convergence theorem (because ϕ^{-1} is bounded on $[0, 1]$), we have that

$$\begin{aligned} & \lim_{\sigma_L \rightarrow \infty} \mathbb{E}_{q_t, \varepsilon_t, \varepsilon'_t \sim \mathcal{N}(0, \sigma_H^2 - \sigma_L^2, \theta, 1)} \left[\phi^{-1} \left(\frac{q_t + \varepsilon_t + \varepsilon'_t}{1 + \sigma_L^2} \right) \right] \\ &= \lim_{\sigma_L \rightarrow \infty} \mathbb{E}_{q_t, \varepsilon_t, \varepsilon'_t \sim \mathcal{N}(0, \sigma_H^2 - \sigma_L^2, 0, \theta)} \left[\phi^{-1} \left(\frac{q_t + \varepsilon_t + \varepsilon'_t}{1 + \sigma_L^2} \right) \right] \\ &= \phi^{-1}(0), \end{aligned}$$

which implies that for any bribe $b^* > 0$, the firm prefers to take action **Not Bribe**. There are two cases. The first is that σ_H is bounded above as $\Delta \rightarrow \infty$. Because $\beta > 0$, $I_\infty(\sigma_H)$ is nonvanishing as $\Delta \rightarrow \infty$, the high type reviewer mimics her own type as $\sigma_L \rightarrow \infty$, which makes the environment bribe proof and therefore attains first-best consumer utility. The second is that $\sigma_H \rightarrow \infty$ in which case the consumption is given by $X_t^* = \phi^{-1}(0)$ regardless of the firm's bribe, so for any $b^* > 0$, the firm's best response is **Not Bribe**. $\square$

Proof of Proposition 3. We show there exists p^* such that the environment is bribe proof when $p < p^*$ and is not bribe proof when $p > p^*$. This establishes the claim because consumer utility in the bribe-proof environment is $p\bar{C}U_\infty(H) + (1-p)\bar{C}U_\infty(L)$ (which is increasing in p), and consumer utility is just $\bar{C}U_\infty(L)$ in an environment that is not bribe proof, with $\bar{C}U_\infty(H) > \bar{C}U_\infty(L)$ by Proposition 1.

The value of bribing for a firm is given by the difference between the firm entering and bribing and its next best option (either entering and not bribing, or staying out). Let $\underline{X}$ and $\bar{X}$ denote expected consumption when the reviewer biases the firm down and up, respectively, when she is

actually the high type reviewer but mimics the low type. If the firm enters and bribes, it receives $p\bar{X} + (1-p)\bar{X}^*$, whereas if the firm enters and does not bribe, it receives $p\underline{X} + (1-p)\bar{X}^*$, and if it stays out it receives γ . Therefore, the value of the bribe is given by

$$\begin{aligned} & \min\{p\bar{X} + (1-p)\bar{X}^* - (p\underline{X} + (1-p)\bar{X}^*), p\bar{X} + (1-p)\bar{X}^* - \gamma\} \\ &= p\bar{X} + \min\{-p\underline{X}, (1-p)\bar{X}^* - \gamma\}. \end{aligned}$$

The environment is bribe proof if $p\bar{X} + \min\{-p\underline{X}, (1-p)\bar{X}^* - \gamma\} < b^* < \beta(I_\infty(\sigma_H) - I_\infty(\sigma_L))$ and otherwise admits bribes (per Theorem 1). In the event that $\beta(I_\infty(\sigma_H) - I_\infty(\sigma_L)) > b^*$, the reviewer never relinquishes her influence for a bribe (regardless of p), so the environment is always bribe proof and taking any $p^* > 1$ establishes the claim.

Otherwise, $b^* < \beta(I_\infty(\sigma_H) - I_\infty(\sigma_L))$, and it is enough to check if $p\bar{X} + \min\{-p\underline{X}, (1-p)\bar{X}^* - \gamma\} < b^*$ to determine whether the environment is bribe proof. Because b^* is constant with respect to p but $p\bar{X} + \min\{-p\underline{X}, (1-p)\bar{X}^* - \gamma\}$ is increasing in p (as $\bar{X} > \bar{X}^* > \underline{X}$), if the inequality holds with a given p , it still holds with any $p' < p$, so the environment is still bribe proof for any $p' < p$ if it is bribe proof with p . By monotonicity, there exists a cutoff p^* where $p < p^*$ is bribe proof but $p > p^*$ is not. $\square$

Proof of Proposition 4. The first-best level of welfare for the consumer was established in Proposition 1. Because ϕ is symmetric around $1/2$, $\phi^{-1}(z) = 1 - \phi^{-1}(-z)$ for $z \geq 0$, or $\phi^{-1}(z) + \phi^{-1}(-z) = 1$ for all $z \geq 0$. Similar to Proposition 1 by strong law of large numbers, we know that, for the firms, the average consumption satisfies

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t} \sum_{\tau=1}^t X_\tau^* &\stackrel{a.s.}{=} \mathbb{E}_{r_t \sim \mathcal{N}(0, 1 + \hat{\sigma}^2)} \left[\phi^{-1} \left(\frac{r_t}{1 + \hat{\sigma}^2} \right) \right] \\ &= \frac{1}{\sqrt{2\pi\hat{\sigma}}} \int_{-\infty}^{\infty} \phi^{-1} \left(\frac{\alpha}{1 + \hat{\sigma}^2} \right) \cdot \exp \left(-\frac{\alpha^2}{2\hat{\sigma}^2} \right) d\alpha \\ &= \frac{1}{\sqrt{2\pi\hat{\sigma}}} \int_0^{\infty} \left(\phi^{-1} \left(\frac{\alpha}{1 + \hat{\sigma}^2} \right) + \phi^{-1} \left(\frac{-\alpha}{1 + \hat{\sigma}^2} \right) \right) \cdot \exp \left(-\frac{\alpha^2}{2\hat{\sigma}^2} \right) d\alpha \\ &= \frac{1}{\sqrt{2\pi\hat{\sigma}}} \int_0^{\infty} \exp \left(-\frac{\alpha^2}{2\hat{\sigma}^2} \right) d\alpha \\ &= 1/2, \end{aligned}$$

independent of $\hat{\sigma}$. Thus, regardless of which type the reviewer mimics, the average consumption of all firms is a constant.

In the expected welfare calculation for the firm, we have $\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{\tau=1}^t X_\tau^*$ when the environment is bribe proof (separating equilibrium). When the environment admits bribes (i.e., pooling equilibrium), firm welfare is given by $\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{\tau=1}^t X_\tau^* - (1-\theta)b^*$, where $\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{\tau=1}^t X_\tau^*$ is the same as before by the previous argument. Because $b^* > 0$, firm welfare strictly decreases with the introduction of bribes.

Any bribes exchanged between the reviewer and the firm offset in the welfare calculation. Moreover, the reviewer obtains the most influence in a bribe-proof environment because she mimics her true type and not a lower type (with less influence). Therefore, the first-best welfare is obtained for the firm and the consumer in the bribe-proof environment (with strictly lower welfare for both in the pooling equilibrium). The bribe-proof

environment always maximizes total welfare: the firm's expected consumption is unchanged, the consumer does worse, the reviewer's influence decreases, and the bribes are neither net positive or negative, because they are simply transfers between agents. $\square$

A.2. Supplemental Results

We show that the strategy the reviewer uses throughout the paper is the one that leads to the highest review bias for bribing firms. Let m be the map introduced in Section 3 for the type H reviewer mimicking type L. A type H reviewer who commits to playing $\hat{\omega} = L$ writes reviews $r_t = s_t + \varepsilon'_t$, where the additional noise ε'_t she injects into her signal is determined in the following way:

- If the firm chooses **Not Bribe**, then the reviewer adds noise $\varepsilon'_t \sim \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, 0, \theta)$
- If the firm chooses **Bribe**, then the reviewer adds noise $\varepsilon'_t \sim \mathcal{N}(0, \sigma_L^2 - \sigma_H^2, \theta, 1)$

Lemma A.1 (Mimicking Lemma). *The difference in expected payoff from consumption (i.e., $\mathbb{E}[X_t^*]$) between actions **Bribe** and **Not Bribe** is maximized under mapping m , subject to the constraint that $\theta \cdot r_t(\text{Not Bribe}) + (1 - \theta) \cdot r_t(\text{Bribe})$ has the same joint distribution with q_t as a truthful type L reviewer (i.e., $(r_t, q_t) \sim \mathcal{N}(\mathbf{0}, \Sigma)$, where $\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 + \sigma_L^2 \end{pmatrix}$).*

Proof of Lemma A.1. Step 1: Let $R: (s_t, b_t) \mapsto r_t$ be any reviewer strategy that maps signals s_t and bribes b_t to reviews r_t and satisfies the distributional constraints; that is, $\tilde{R}(s_t) \equiv \theta R(s_t, 0) + (1 - \theta)R(s_t, b^*)$ is constrained to be jointly normal with q_t , satisfying (1) $\mathcal{N}(0, 1 + \hat{\sigma}^2)$ and (2) $\text{cov}(\tilde{R}(s_t), q_t) = 1$. Constraint (1) ensures the reviewer matches a truthful reviewer of some type $\hat{\omega}$ and Condition (2) ensures it has the proper covariance with q_t (i.e., jointly normal variables with correlation $1/\sqrt{1 + \hat{\sigma}^2}$). Because the consumers do not observe the bribes b_t , given that θ firms bribe 0 and $(1 - \theta)$ firms bribe b^* , $\tilde{R}$ is the distribution that consumers observe over time. Although consumers do not observe signals either, they can measure how $\tilde{R}$ compares to a truthful reviewer who trivially maps signals $s_t \mapsto R_t$ (but ignores any b_t).

First, note that $\tilde{R}(s_t) - s_t$ is normally distributed: It is the difference of two jointly normal random variables. Similarly, observe that $\text{cov}(\tilde{R}(s_t) - s_t, q_t) = \text{cov}(\tilde{R}(s_t), q_t) - \text{cov}(s_t, q_t) = 1 - 1 = 0$; therefore, $\tilde{R}(s_t) - s_t$ is uncorrelated with q_t . Because both are normally distributed, this implies that they are in fact independent as well.

Hence, for any strategy $\tilde{R}(s_t) - s_t$ to be independent of q_t , it must be the case that $\tilde{R}(s_t) - s_t$ is independent of s_t , because all of these are jointly normal, but s_t and q_t are correlated. In summary, $\tilde{R}(s_t) - s_t$ must have the distribution $\mathcal{N}(0, \hat{\sigma}^2 - \sigma_\omega^2)$ and be orthogonal to s_t , for any choice of R for the reviewer. Therefore, we can write $\tilde{R}(s_t) = s_t + \varepsilon'_t$, where ε'_t is orthogonal to s_t and has distribution $\mathcal{N}(0, \hat{\sigma}^2 - \sigma_\omega^2)$. Rewritten,

$$\tilde{R}(s_t) = s_t + \theta(R(s_t, 0) - s_t) + (1 - \theta)(R(s_t, b^*) - s_t).$$

Step 2: Next, we consider the map $m(s_t, 0) = s_t + \varepsilon'_t$ with $\varepsilon'_t \sim \mathcal{N}(0, \hat{\sigma}^2 - \sigma_\omega^2, 0, \theta)$ and $m(s_t, b^*) = s_t + \varepsilon'_t$ with $\varepsilon'_t \sim \mathcal{N}(0, \hat{\sigma}^2 -$

$\sigma_\omega^2, \theta, 1)$ (and $\tilde{m}(s_t) \equiv \theta m(s_t, 0) + (1 - \theta)m(s_t, b^*)$). It is easy to see $\tilde{m}$ satisfies the required distributional constraint (1):

$$\mathbb{P}[\tilde{m}(s_t) \leq a] = \begin{cases} \theta \mathbb{P}[m(s_t, 0) \leq a], & \text{if } a \leq \sqrt{\hat{\sigma}^2 - \sigma_\omega^2} \Phi^{-1}(\theta) \\ \theta + (1 - \theta) \mathbb{P}[m(s_t, b^*) \leq a] & \text{if } a \geq \sqrt{\hat{\sigma}^2 - \sigma_\omega^2} \Phi^{-1}(\theta). \end{cases}$$

Note that $\mathbb{P}[m(s_t, 0) \leq a] = \frac{1}{\theta} \mathbb{P}[\mathcal{N}(0, \hat{\sigma}^2 - \sigma_\omega^2) \leq a]$ when $a \leq \sqrt{\hat{\sigma}^2 - \sigma_\omega^2} \Phi^{-1}(\theta)$ and $\mathbb{P}[m(s_t, b^*) \leq a] = \frac{1}{1 - \theta} (\mathbb{P}[\mathcal{N}(0, \hat{\sigma}^2 - \sigma_\omega^2) \leq a] - \theta)$ when $a \geq \sqrt{\hat{\sigma}^2 - \sigma_\omega^2} \Phi^{-1}(\theta)$. Thus, $\mathbb{P}[\tilde{m}(s_t) \leq a] = \mathbb{P}[\mathcal{N}(0, \hat{\sigma}^2 - \sigma_\omega^2) \leq a]$. It is also trivially satisfies distributional constraint (2) because ε'_t is drawn independently from s_t (and q_t).

Suppose that $m(s_t, b^*) - s_t$ first-order stochastically dominates all other distributions $R(s_t, b^*) - s_t$ subject to the distributional constraints that $\tilde{R}$ satisfy: (i) $\tilde{R}(s_t) - s_t$ is orthogonal to s_t ; (ii) $\tilde{R}(s_t) - s_t \sim \mathcal{N}(0, \hat{\sigma}^2 - \sigma_\omega^2)$. This also implies that $m(s_t, 0)$ is first-order stochastically dominated by all other $R(s_t, 0) - s_t$ given distributional restrictions (i) and (ii). This means that $\mathbb{E}[X_t^* | b_t = b^*]$ under the reviewer strategy of m is larger than under any other strategy R that satisfies the constraints and $\mathbb{E}[X_t^* | b_t = 0]$ under the reviewer strategy of m is smaller than under any other strategy R satisfying the constraints. Thus, their difference is maximized amongst all such R with mapping m .

Step 3: It remains to prove first-order stochastic dominance of $m(s_t, b^*) - s_t$ over any other $R(s_t, b^*) - s_t$, as we do next. Consider some arbitrary $\tilde{R}(s_t)$ satisfying conditions (1) and (2) as before (where recall $\tilde{R}(s_t) = \theta R(s_t, 0) + (1 - \theta)R(s_t, b^*)$). For any $a < (\hat{\sigma}^2 - \sigma_\omega^2) \Phi^{-1}(1 - \theta)$, then $\mathbb{P}[m(s_t, b^*) - s_t \geq a] = 1$ (i.e., any a that is below the $(1 - \theta)$ th percentile of the normal distribution with variance $\hat{\sigma}^2 - \sigma_\omega^2$, $\varepsilon'_t = m(s_t, b^*) - s_t$ is necessarily drawn above a , by construction). Hence, $\mathbb{P}[m(s_t, b^*) - s_t \geq a] \geq \mathbb{P}[R(s_t, b^*) - s_t \geq a]$ trivially, because the latter is a probability measure between zero and one.

Conversely, for any $a \geq \sqrt{\hat{\sigma}^2 - \sigma_\omega^2} \Phi^{-1}(1 - \theta)$, note that

$$\mathbb{P}[m(s_t, b^*) - s_t \geq a] = \frac{1 - \Phi\left(\frac{a}{\sqrt{\hat{\sigma}^2 - \sigma_\omega^2}}\right)}{1 - \theta}$$

as the bottom $(1 - \theta)$ fraction of the distribution occurs with a probability of zero, and the top $(1 - \theta)$ fraction of the distribution is drawn proportional to the normal density, so the tail end of the distribution is amplified by $1/(1 - \theta)$. Because of distributional constraints (i) and (ii) for $\tilde{R}(s_t) - s_t$, we know that $1 - \Phi\left(\frac{a}{\sqrt{\hat{\sigma}^2 - \sigma_\omega^2}}\right) = \mathbb{P}[\tilde{R}(s_t) - s_t \geq a]$; thus,

$$\begin{aligned} &= \frac{1}{1 - \theta} \mathbb{P}[\tilde{R}(s_t) - s_t \geq a] \\ &= \frac{1}{1 - \theta} (\theta \mathbb{P}[\tilde{R}(s_t) - s_t \geq a | b_t = 0] + (1 - \theta) \mathbb{P}[\tilde{R}(s_t) - s_t \geq a | b_t = b^*]) \\ &= \frac{\theta}{1 - \theta} \mathbb{P}[\tilde{R}(s_t) - s_t \geq a | b_t = 0] + \mathbb{P}[\tilde{R}(s_t) - s_t \geq a | b_t = b^*] \\ &\geq \mathbb{P}[\tilde{R}(s_t) - s_t \geq a | b_t = b^*] \\ &= \mathbb{P}[R(s_t, b^*) - s_t \geq a], \end{aligned}$$

where the second equality follows from the law of total probability, the third equality distributes the fraction $1/(1 - \theta)$, the

inequality follows from the fact that $\frac{\theta}{1-\theta} \mathbb{P}[\tilde{R}(s_t) - s_t \geq a | b_t = 0] \geq 0$ (i.e., probabilities are nonnegative), and the final equality is by definition. This establishes that $m(s_t, b^*) - s_t$ first-order stochastically dominates *any* arbitrary reviewer strategy $R(s_t, b^*) - s_t$, satisfying the distributional constraints required, which completes the proof of Lemma A.1. $\square$

Endnotes

¹ Some of this literature (Candogan and Drakopoulos 2020) also touches on the larger issues of information disclosure, as discussed in Ottaviani and Sørensen (2006) or Dye (1985).

² Quoting Durbin and Iyer (2009, p. 224): “our advisor does not care about a reputation for accuracy, but rather a reputation for incorruptibility ... in general, these two types of reputational incentives do not coincide.”

³ Here, b^* is taken as exogenous, but an endogenous choice of b_t (over a continuous action space) does not affect the main insights of the model, as explained informally in Section 3.5 and formally shown in Online Appendix B.

⁴ The assumption that a reviewer can mimic lower types but not higher types is reminiscent of the reputation papers of Milgrom (1981) and Grossman (1981). We show the connection formally as an outcome of the fully dynamic model in Lemma B.7, stated and proven in Online Appendix B. We discuss this connection informally in Section 3.5.

⁵ Formally, r_t must be statistically identical to one that is drawn from $r_t = q_t + \varepsilon_t$, with ε_t independent of q_t and distributed as $\mathcal{N}(0, \sigma_\varepsilon^2)$, given that q_t is observed at the end of each period and can be compared with r_t .

⁶ This is shown formally in Lemma A.1.

⁷ Equivalently, we can assume that consumers, like the reviewer, get a noisy experience that is centered around the true quality; that is, they get experience $e_{i,t} = q_t + \eta_{i,t}$ for some unbiased noise term $\eta_{i,t}$. The noise in the consumer’s experience allows for the possibility that consumers do not all have identical ex post utility from using the product. This does not change any of our results.

⁸ By “generic conditions,” we mean that for the set of initial parameters in the model, the subset where there are exceptions is a null set, that is, has measure zero. Therefore, although other knife-edge cases might exist, perturbing the parameters by an arbitrarily small amount is guaranteed to eliminate these exceptions. Hence, these nongeneric cases are not stable outcomes of the model and thus will not be a primary focus.

⁹ This is a classic signal extraction problem (Cochran 1954); take $Z = W + Y$, where $W \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, \sigma_Y^2)$. Then $\mathbb{E}[W | Z = z] = \frac{z/\sigma_Y^2}{1 + 1/\sigma_Y^2}$ (i.e., the prior of $\mathcal{N}(0, 1)$ regresses the estimate of W closer to zero).

¹⁰ See, for example, section 1.C in Kamenica and Gentzkow (2011), where the reviewer in our paper is the “Sender” and each consumer is a “Receiver.”

¹¹ Equivalently, one can measure total utility by summing over the outside option of inactive consumers and adding $q_t \cdot \mu(\mathcal{A}_t)$ (i.e., the utility of active consumers). These two quantities differ only by the constant $\int_0^1 \phi_i di$.

¹² Even if that was not the case, a more complex (but equivalent) model guarantees that some nonbribing firms will still enter. In an effort to not overcomplicate our model, we abstract away from the detail that firms may have *heterogeneous* outside options. This makes some nonbribing firms opt out of entering if they believe they will

get reviews that are not biased in their favor but still permits those with lower outside options to enter. We incorporate this into our model by assuming that the proportion θ is already calibrated so that it designates only those (nonbribing) firms that still find it profitable to enter and not bribe, regardless of whether these firms expect a bad review.

¹³ We provide intuition for the discussion in this paragraph in Appendices B.5.4 and B.5.5 of Online Appendix B.5.

¹⁴ Because the consumer cannot observe the bribes, this distribution depends only on the effective type $\hat{\omega}$ (with corresponding variance $\hat{\sigma}^2$) and not on the true type ω (or the true signal variance σ_ω^2). This is shown in Lemma A.1 in Appendix A.2.

¹⁵ By total mixing, we mean that the firm chooses each action with positive probability.

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